

Weight of Grade: 40% of grade

Due Date:

LIAI 1009 - Deep Learning

Development and Application of a Generative Adversarial Network

Objective: Develop a GAN to generate new data in a domain of choice, analyze its performance, and explore its creative and practical applications.

Overview: Students will create a GAN that generates realistic images, text, or other forms of media. They will then assess the quality of the generated data, optimize the GAN architecture, and explore potential applications of their model.

Tasks:

1. Choice of Application:

- Choose a specific domain for data generation, such as digital art, fashion design, realistic human faces, new levels for video games, or synthetic medical images for training diagnostics.

2. GAN Model Development:

- Implement a GAN model from scratch or modify an existing architecture.
- Use a suitable dataset to train the GAN. For image generation, consider datasets like CIFAR-10, CelebA, or a custom dataset relevant to the chosen application.

3. Training the GAN:

- Train the GAN, focusing on the interplay between the generator and discriminator to achieve realistic outputs.
- Experiment with different hyperparameters, architectures, and training techniques to optimize the generation process.

4. Quality Assessment:

- Evaluate the quality of the generated data using both qualitative methods (visual inspection) and quantitative methods (inception score, FID).
- Document how modifications to the model architecture or training process affect the output quality.

5. **Creative and Practical Applications:**

- Explore and propose potential creative or practical applications of the generated data within the chosen domain.
- Develop a mini-project or prototype that utilizes the generated data, demonstrating its potential real-world application.

6. **Ethical Considerations:**

- Discuss the ethical implications of using GANs in the chosen domain, including issues around data privacy, the potential for generating deceptive media, and the impact on jobs in creative industries.

7. **Report and Presentation:**

- Compile a comprehensive report detailing the development process, experiments, results, and conclusions.
- Prepare a presentation summarizing the project findings, showcasing examples of generated outputs, and discussing potential real-world applications.

Submission Requirements:

- Complete source code of the GAN implementation along with instructions on how to run it.
- A detailed project report including methodology, results, and discussion on the effectiveness and applications of the GAN.
- A presentation that can be shared with the class during a final project showcase.

